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EEEM068: Applied Machine Learning
Project: Video Anomaly Detection by Weakly Supervised Methods

- Submission deadline: **TBC**

Project Specification

Overview

This project requires students to apply the theoretical knowledge and practical skills acquired in the EEEM068: Applied Machine Learning module. The project contributes 100% of the final module mark and is assessed as follows: 60% for the submitted code and written report, and 40% for the oral examination (viva).

The objective of this project is to design, implement, and evaluate a deep learning framework for **Video Anomaly Detection (VAD)**. Students are expected to develop a model capable of identifying abnormal events in surveillance videos and to critically analyse its performance.

Submission

You must submit a single compressed (.zip) file containing:

- Your complete, well-structured, and fully documented source code;
- A **5-page project report** written in IEEE double-column format.

The report must clearly describe:

- The problem formulation;
- The chosen methodology and model architecture;
- Implementation details;
- Experimental setup;
- Quantitative and qualitative results;
- Critical discussion of findings.

You may include additional pages containing only visualisations (figures with short captions). These extra pages are optional and will not affect marking. However, the main body of the report must not exceed four pages.

All figures and tables must be properly numbered and referenced within the main text.

Background

Video surveillance is widely used to monitor environments such as public spaces, transportation systems, and private property. However, manual monitoring of video streams is time-consuming, labour-intensive, and prone to human error. Automated analysis using machine learning and computer vision techniques offers scalable and efficient solutions.
Video Anomaly Detection (VAD) aims to identify frames or temporal segments that deviate from normal patterns of behaviour [1]. It plays a critical role in public safety [3] and intelligent video understanding [5]. Typical anomalies may include events such as accidents, theft, violence, explosions, or other unusual activities.
A major challenge in VAD is the scarcity of precisely annotated data. Weakly supervised approaches [3, 4, 2, 6] address this issue by using video-level labels instead of frame-level annotations, significantly reducing labelling costs while maintaining strong performance. Multiple Instance Learning (MIL), introduced for anomaly detection by Sultani *et al.* [3], has become a dominant paradigm in weakly supervised VAD. In this framework, each video is treated as a bag of temporal segments, and the model learns to assign anomaly scores to individual segments.

For this project, students will use the UCF-Crime dataset [3] and XD Violence[6] Dataset to train and evaluate their models. Different modelling approaches may be explored, including Convolutional Neural Networks (CNNs), 3D CNNs, recurrent architectures, or Vision Transformers like CLIP-based models. Students are expected to justify their design choices and critically evaluate model performance using appropriate metrics.

Project Steps

- **Dataset Preparation:** Download and organise the UCF-Crime dataset and XD-Violence dataset. Extract video features (e.g., using a pre-trained CNN such as I3D, C3D, or ResNet). Create a structured dataset where each video is divided into temporal segments and assigned a video-level label (normal or anomalous).
- **Data Splitting:** Split the dataset into training, validation, and testing sets (e.g., 80% training, 10% validation, 10% testing), ensuring no data leakage between splits.
- **Feature Representation:** Preprocess the extracted features and organise them into fixed-length segment representations suitable for Multiple Instance Learning (MIL). Apply appropriate normalisation if required.
- **Model Design (Weakly Supervised Setting):** Implement a weakly supervised Video Anomaly Detection model using a Multiple Instance Learning (MIL) framework. The model should take a bag of temporal segments as input and produce anomaly scores per segment. Focus of developing a spatiotemporal attention-based model that will effectively learn both the temporal and the spatial info from input features. They try to design a loss function that would effectively help the model to discriminate between the Normal and the abnormal features. Make sure the model is able to process the input in a real time . If possible show the comparative comparisons interms of the computational complexity like space and time complexity(infernence time) .

  Students may:

  - Implement a baseline MIL model as proposed by Sultani et al.;
  - Extend the baseline with temporal attention mechanisms;
  - Experiment with alternative architectures such as 1D temporal CNNs, LSTMs, or Transformers.
- **Loss Function:** Implement an appropriate weakly supervised loss function (e.g., ranking loss between normal and anomalous videos) that encourages higher anomaly scores for abnormal segments.
- **Model Training:** Train the model using the training set. Carefully tune hyperparameters such as learning rate, batch size, number of segments per video, and regularisation strategies.
- **Model Evaluation:** Evaluate performance on the test set using appropriate metrics for anomaly detection, such as:

  - Area Under the ROC Curve (AUC);
  - Frame-level ROC analysis;
  - Precision-Recall curves.
  - inference time or speed.
- **Visualisation and Analysis:** Plot anomaly score curves over time for selected videos. Visualise correctly detected anomalies as well as failure cases. Plot UMAP and TSNE visualizations of the abnormal vs normal features based on the loss function. Provide a critical analysis explaining:

  - Why certain anomalies are correctly detected;
  - Why some normal segments may be misclassified;
  - The limitations of weak supervision.
- **Discussion and Improvements:** Propose and experimentally evaluate at least one improvement over the baseline (e.g., feature fusion, attention mechanism, temporal smoothing, or data balancing strategy,Loss function development).

Extra credit

You can integrate the textual space with the developed weakly supervised model by integrating the Clip models' textual encoder as an addition, so that the model learn the semantic space as well, thus at the output model not only generates the anomaly scores but also can generate the corresponding anomaly textual label. Compare your model's performance against that of current weakly supervised VAD models and see if you can fine-tune to improve your performance. You could also have a look at visualisation, which I have a very basic example of in my GitHub (https://github.com/Apthy/VisualisingNNs)

Marking Criteria

The assessment for this project will be divided into two main components:

- **Group Coursework Submission (60%)**: Includes the technical report and working code.
- **Group Presentation and Oral Exam/Viva (40%)**: The presentation will be 5 minutes long, followed by a 15-minute Q&A session with questions directed at each group member.

Group Requirements

- Each group should consist of **four/five students**.
- **Equal Contribution**: All members must contribute equally to:
  - Technical/coding tasks.
  - Experiments.
  - Report writing.
  - Presentation.
- The project should utilize **PyTorch**, **Python notebooks**, and **Google Colab**. Access to the **Surrey AI Supercomputer** is also available.

Assessments: Marking Criterion

The group submission aspect of the project will be assessed as follows:

- **Technical Report (50% - 30 marks)**: Clarity, structure, results presentation, and discussion.
- **Functionality (30% - 18 marks)**: Completeness of implementation and performance of the model.
- **Code Quality (20% - 12 marks)**: Efficiency, maintainability, and thorough documentation.

Detailed Report Assessment (30 marks)

- Abstract: 5 marks
- Introduction: 5 marks
- Literature Review (minimum 5 papers): 5 marks
- Methodology: 7 marks
- Experiments: 5 marks
- Conclusion and Future Work: 3 marks

Functionality (18 marks)

- Successful implementation of all required steps, with thorough validation of results.

Code Quality (12 marks)

- Efficient, maintainable, and well-documented code adhering to best practices.

References

[1] Mahmudul Hasan, Jonghyun Choi, Jan Neumann, Amit K. Roy-Chowdhury, and Larry S. Davis. Learning temporal regularity in video sequences, 2016.

[2] Hui Lv, Zhongqi Yue, Qianru Sun, Bin Luo, Zhen Cui, and Hanwang Zhang. Unbiased multiple instance learning for weakly supervised video anomaly detection, 2023.

[3] Waqas Sultani, Chen Chen, and Mubarak Shah. Real-world anomaly detection in surveillance videos, 2019.

[4] Yu Tian, Guansong Pang, Yuanhong Chen, Rajvinder Singh, Johan W. Verjans, and Gustavo Carneiro. Weakly-supervised video anomaly detection with robust temporal feature magnitude learning, 2021.

[5] Peng Wu, Jing Liu, Yujia Shi, Yujia Sun, Fangtao Shao, Zhaoyang Wu, and Zhiwei Yang. Not only look, but also listen: Learning multimodal violence detection under weak supervision, 2020.

[6] Peng Wu, jing Liu, Yujia Shi, Yujia Sun, Fangtao Shao, Zhaoyang Wu, and Zhiwei Yang. Not only look, but also listen: Learning multimodal violence detection under weak supervision. In *European Conference on Computer Vision (ECCV)*, 2020.